

Chapter 2: Network and Protocol Fundamentals

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Duration: 90 minutes

Goal: Understand network structures, core protocols, and how their weaknesses influence network security — forming the basis for traffic analysis.

Lecture Overview

1. What is a Computer Network?
2. Network Models (TCP/IP vs. OSI)
3. Key Protocols: HTTP, HTTPS, ARP, DNS
4. Common Network Devices
5. Protocol Weaknesses and Security Implications
6. Foundation for Traffic Analysis

1. What is a Computer Network?

Definition:

A system that connects computers and devices to share resources, exchange data, and communicate efficiently.

Core Functions:

- Data communication
- Resource sharing
- Centralized management
- Security and monitoring

Examples:

LAN (Local Area Network), WAN (Wide Area Network), Internet

Why Study Network Fundamentals?

- Every cyber attack traverses a **network path**
- Understanding **how data flows** → ability to **detect anomalies**
- Forms the **technical basis of cybersecurity monitoring**

Instructor Note:

Before defending the network, we must understand its behavior.

1.5 Basic Network Addresses Required for Internet Connectivity

The Four Essential Addresses

To connect any device to the Internet, four fundamental network parameters must be configured:

1. IP Address
2. Subnet Mask (Netmask)
3. Default Gateway
4. DNS Server

These define the **identity, boundary, route, and naming resolution** for all network communications.

IP Address

Definition:

A unique logical identifier assigned to each device on a network.

Types:

- **IPv4:** 32-bit (e.g., 192.168.1.10)
- **IPv6:** 128-bit (e.g., 2001:db8::1)

IP Address

Purpose:

- Uniquely identify the host on the network
- Enable routing between different networks

Example:

A laptop with IP `192.168.1.10` can communicate within its subnet and reach the Internet via its gateway.

Subnet Mask (Netmask)

Definition:

Determines which portion of the IP address represents the network and which part identifies the host.

Example:

- IP: 192.168.1.10
- Subnet Mask: 255.255.255.0
→ Network: 192.168.1.0/24

Subnet Mask (Netmask)

Purpose:

- Defines local communication boundaries
- Ensures correct packet delivery within a LAN

Security Note:

Incorrect subnet configuration can expose devices or break segmentation.

Default Gateway

Definition:

The device (usually a router) that forwards traffic from a local network to other networks, including the Internet.

Example:

- Gateway: 192.168.1.1
- Function: Sends packets to external destinations not in the local subnet.

Role in Security:

- Acts as the **first line of defense** for outbound traffic
- Can apply routing and firewall rules

DNS Server (Domain Name System)

Definition:

Translates human-readable domain names into IP addresses.

Example:

- Query: `www.wu.ac.th` → Response: `203.158.xxx.xxx`

Function:

- Enables convenient access to online services
- Reduces user error from IP memorization

Security Consideration:

- DNS spoofing or poisoning can redirect users to malicious sites.
- Secure alternatives: **DNSSEC, DNS over HTTPS (DoH).**

Visualization: Network Parameter Relationship

```
[Device]
↓ (IP Address)
[Local Network]
↓ (Subnet Mask defines boundary)
[Gateway Router]
↓ (Route to Internet)
[DNS Server]
↓ (Resolves domain names)
[Global Internet]
```

Summary:

All four parameters work together to create **network identity, scope, and reachability** — forming the basis of all Internet communication.

2. Network Models Overview

Two major conceptual models guide communication:

Model	Layers	Example
OSI Model	7 Layers (Physical → Application)	Educational reference model
TCP/IP Model	4 Layers (Network Access, Internet, Transport, Application)	Practical Internet model

Insight:

Most modern systems use the **TCP/IP model**, a simplified implementation of OSI concepts.

OSI Model — The 7 Layers

1. **Physical** – Hardware transmission
2. **Data Link** – MAC addressing, frames
3. **Network** – Routing (IP)
4. **Transport** – Reliable delivery (TCP/UDP)
5. **Session** – Connection management
6. **Presentation** – Data format (encryption, compression)
7. **Application** – End-user protocols (HTTP, DNS, SMTP)

TCP/IP Model — The Practical Stack

1. Network Access Layer
2. Internet Layer (IP)
3. Transport Layer (TCP/UDP)
4. Application Layer

Mapping:

- OSI Layers 1–2 → TCP/IP Network Access
- OSI 3 → Internet
- OSI 4 → Transport
- OSI 5–7 → Application

TCP/IP Communication Flow

Process Example:

User opens `https://example.com`

1. Application: Browser sends HTTP request
2. Transport: TCP segments data
3. Internet: IP routes packets
4. Network Access: NIC transmits over Ethernet/Wi-Fi

Result: Seamless communication between applications worldwide.

3. Key Protocols in Focus

We focus on foundational Internet protocols that define data exchange:

- **HTTP / HTTPS** — Web communication
- **ARP** — Address Resolution
- **DNS** — Name resolution

HTTP (Hypertext Transfer Protocol)

Function:

Transfers data between web clients and servers (default port 80).

Characteristics:

- Plain text communication
- Stateless
- Readable by anyone capturing packets

Security Issue:

Lacks encryption → **susceptible to sniffing and man-in-the-middle attacks.**

HTTPS (HTTP Secure)

Enhancement:

HTTP over **TLS/SSL** (default port 443)

Benefits:

- Data encryption
- Server authentication
- Message integrity

Example:

`https://university.ac.th` ensures data privacy between student and server.

Security Limitation:

Improper certificate management can still expose users to **phishing or downgrade attacks**.

DNS (Domain Name System)

Purpose:

Translates domain names into IP addresses.

e.g., `www.google.com` → `142.250.190.14`

Mechanism:

Hierarchical lookup through recursive resolvers.

Weaknesses:

- Traditionally uses **UDP** (no encryption)
- Susceptible to **DNS spoofing** or **cache poisoning**
- Modern improvement: **DNS over HTTPS (DoH)** or **DNSSEC**

Transmission Control Protocol (TCP)

Purpose:

Provides reliable, ordered, and error-checked delivery of data between applications.

Key Features:

- Connection-oriented (requires handshake)
- Guarantees delivery
- Performs flow and congestion control

Transmission Control Protocol (TCP)

Process:

1. Three-Way Handshake:

- SYN → SYN/ACK → ACK

2. Data Transfer:

- Reliable stream communication

3. Connection Termination:

- FIN / ACK exchange

Transmission Control Protocol (TCP)

Example:

Used in HTTP, HTTPS, FTP, and SMTP.

Weakness:

- Session hijacking possible if sequence numbers are predictable
- SYN flood → **Denial of Service (DoS)**

User Datagram Protocol (UDP)

Purpose:

Provides fast, connectionless data transmission with minimal overhead.

Characteristics:

- Unreliable (no acknowledgment or retransmission)
- No ordering guarantees
- Lightweight and fast

User Datagram Protocol (UDP)

Examples:

- DNS queries
- Streaming (VoIP, video)
- Online gaming

Weakness:

- Spoofing and amplification attacks (e.g., **UDP flood**)
- Lacks session tracking for authentication or integrity

Comparison Summary

Feature	TCP	UDP
Connection	Yes	No
Reliability	High	Low
Speed	Moderate	Fast
Example	HTTPS, FTP	DNS, VoIP

Internet Protocol (IP)

Purpose:

Handles logical addressing and routing of packets across networks.

Versions:

- **IPv4:** 32-bit address (e.g., 192.168.1.1)
- **IPv6:** 128-bit address (e.g., 2001:db8::1)

Functions:

- Fragmentation and reassembly
- Routing via routers
- Differentiated services (QoS fields)

Internet Protocol (IP)

Weakness:

- No built-in authentication or encryption
- IP spoofing possible — attacker forges source address

Example:

Every Internet communication relies on IP to deliver packets to their destination.

Internet Control Message Protocol (ICMP)

Purpose:

Used by network devices to send control and diagnostic messages.

Common Messages:

- Echo Request/Reply (Ping)
- Destination Unreachable
- Time Exceeded (Traceroute)

Internet Control Message Protocol (ICMP)

Role in Security:

- Helps detect network reachability
- Supports troubleshooting

Weaknesses:

- Can be exploited for **reconnaissance** (ping sweeps)
- Can carry covert data (ICMP tunneling)
- Basis for **ICMP flood (Smurf attack)**

ARP (Address Resolution Protocol)

Purpose:

Maps an IP address to a MAC address within a local network.

Example:

192.168.1.10 → 00:1A:2B:3C:4D:5E

Operation:

Broadcasts ARP requests and replies within LAN.

Weakness:

- No authentication or verification
- Vulnerable to **ARP spoofing/poisoning**, enabling MITM (Man-in-the-middle) attacks.

Summary: TCP/IP Layer Security Risks

Layer	Protocol	Weakness	Common Attack
Transport	TCP	Connection hijacking	SYN Flood
Transport	UDP	Spoofing, flooding	UDP Amplification
Internet	IP	No authentication	IP Spoofing
Internet	ICMP	Diagnostic abuse	Ping flood, ICMP tunneling

Instructor Note:

Understanding these vulnerabilities is essential before examining **network devices** and **traffic analysis tools**.

4. Common Network Devices

Device	Function	Security Role
Router	Routes packets between networks	Controls traffic paths and enforces network boundaries
Switch	Connects devices within a LAN	Enables segmentation (VLANs) and prevents broadcast storms
Firewall	Filters traffic based on rules	Enforces security policies and blocks malicious connections
IDS/IPS	Detects or prevents intrusions	Core of threat detection and early warning systems

4. Common Network Devices

Device	Function	Security Role
Access Point (AP)	Provides wireless network connectivity	Requires strong encryption (WPA3) to prevent eavesdropping
Load Balancer	Distributes network traffic across multiple servers	Increases availability and mitigates overload attacks
Proxy Server	Intermediary for requests between clients and servers	Masks internal addresses and filters web access
VPN Gateway	Creates secure tunnels over public networks	Ensures confidentiality and integrity of remote access

4. Common Network Devices

Device	Function	Security Role
SIEM System	Aggregates logs from multiple devices	Correlates data for real-time security analysis
Network Tap / SPAN Port	Copies network traffic for analysis	Enables passive monitoring for IDS and forensics

Network Devices in Security Architecture

Layered Defense (Defense-in-Depth):

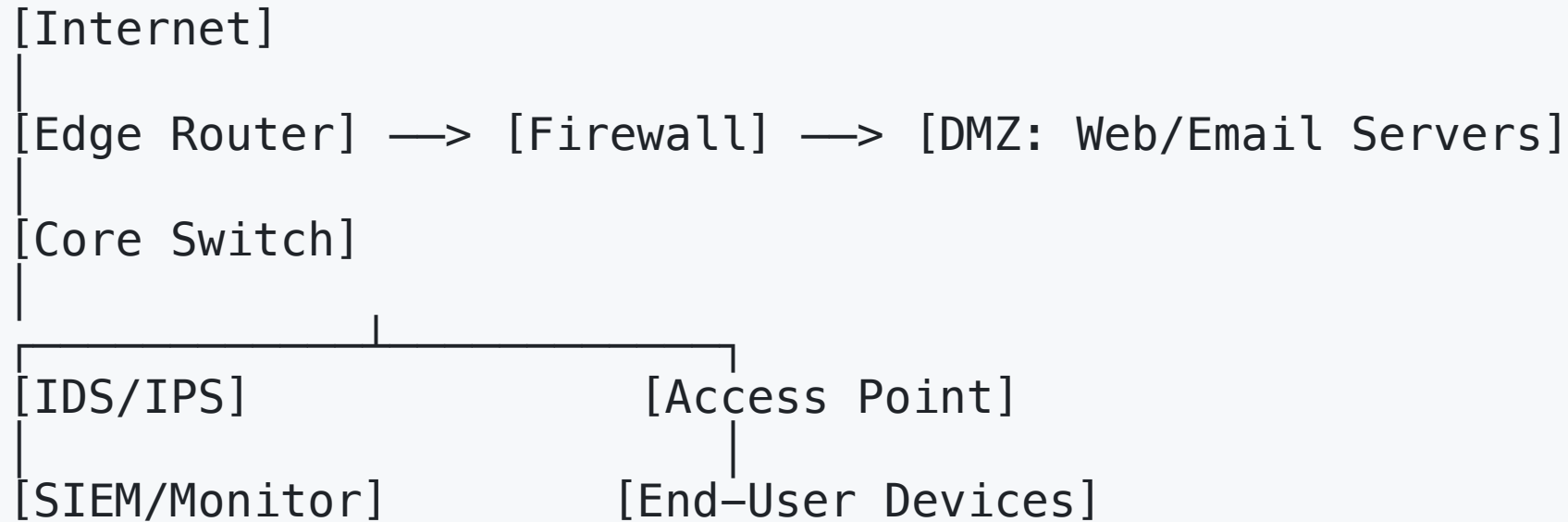
1. **Edge Router** → Manages external connections and applies access control lists (ACLs) to limit exposure.
2. **Firewall** → Examines packets and enforces inbound/outbound policies based on protocols, ports, and IPs.
3. **DMZ (Demilitarized Zone)** → Hosts public-facing services such as web or email servers, isolated from internal LAN.
4. **IDS/IPS** → Monitors internal and external traffic to detect or block suspicious activities.

Network Devices in Security Architecture

Layered Defense (Defense-in-Depth):

- 5. **SIEM / NMS (Security Information & Event Management / Network Management System)** → Aggregates and correlates logs for anomaly detection and compliance.
- 6. **Access Points and VPNs** → Manage wireless and remote entry points, requiring strong authentication.
- 7. **Endpoint Security** → Extends protection to user devices, enforcing antivirus, EDR, and patch management.

Visualizing Network Device Placement



Device Security Considerations

Device	Potential Weakness	Mitigation
Router	Default passwords, outdated firmware	Change credentials, patch regularly
Switch	ARP spoofing, VLAN hopping	Use port security and VLAN isolation
Firewall	Misconfigured rules	Conduct rule reviews and audits
IDS/IPS	False positives or evasion	Tune signatures and correlate with SIEM

Device Security Considerations

Device	Potential Weakness	Mitigation
Access Point	Weak encryption, rogue APs	Enforce WPA3, monitor for rogue devices
VPN Gateway	Key leakage, configuration flaws	Use strong encryption and MFA
Proxy Server	Data caching exposure	Secure cache and restrict admin access

Insight

Each network device:

- **Defines a control point** in the flow of information,
- **Provides logs** crucial for security monitoring, and
- **Supports forensic analysis** when incidents occur.

Key Concept:

A secure network architecture is **not built on a single device**, but through **integration and visibility across all layers** — forming the technical foundation for **traffic analysis** and **intrusion detection** in the next lecture.

5. Traditional Protocol Weaknesses

Protocol	Weakness	Attack Example
HTTP	Unencrypted text	Credential theft
ARP	No authentication	ARP poisoning
DNS	Cache poisoning	Redirect to fake sites
Telnet	Plain-text login	Password sniffing
FTP	Unsecured file transfer	Data interception

Concept:

These weaknesses shaped the development of **secure versions** like HTTPS, SSH, and SFTP.

Visualization: Protocol Evolution

Plain Protocol → Secure Replacement

- HTTP → HTTPS
- Telnet → SSH
- FTP → SFTP / FTPS
- DNS → DNSSEC / DoH

Lesson:

Security evolves through encryption, authentication, and validation.

Protocol Weaknesses and Network Security

Key Insight:

Every packet reveals potential security gaps.

Understanding protocol behavior helps analysts:

- Recognize abnormal flows
- Detect spoofing attempts
- Identify malicious payloads

Summary of Key Concepts

- Networks enable communication but also expand attack surfaces.
- The **TCP/IP model** governs data flow across systems.
- Traditional protocols (HTTP, ARP, DNS) have inherent **security weaknesses**.

- Understanding these is essential for **network defense and monitoring**

Reflection & Discussion

- Why is ARP spoofing still effective in modern networks?
- How can encryption both protect and hinder network monitoring?
- Which protocol's weakness would you prioritize fixing and why?

Thank You

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